QUENCH.

QUANTIFIED EVIDENCE NAVIGATOR FOR COMBUSTION HAZARDS

Team [TEAM NAME] · Members [NAMES OF TEAM MEMBERS]

NASA expects Moon-base air with **30 % oxygen or more**. NASA's open combustion archive holds no solid-fuel flight test above **22.2 %**.

22.2 %
highest oxygen of any solid-fuel flight test in the archive

30 %+
oxygen NASA expects on the Moon, at 10.2 psi or less

0
solid-fuel runs at 30 % oxygen or more

986
flight test runs read from 24 NASA investigations

01

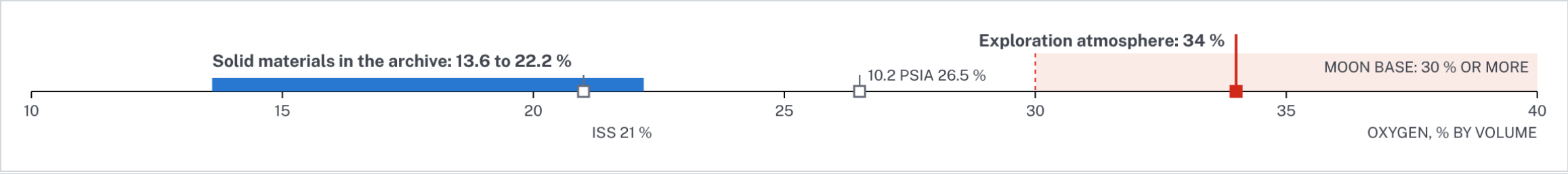
Decades of fire experiments in orbit. Do they cover the air a Moon base will use?

The challenge: a dashboard that **summarizes, ranks and interprets** NASA's microgravity combustion investigations for fire safety.

We read all of them in NASA's PSI archive: **24 investigations, 986 flight test runs**, 352 papers, 6.3 TB of raw data, and placed every run by the oxygen and pressure it burned in.

> "The intermediate oxygen concentration used on the Moon will likely be 30% or greater, with a corresponding pressure of 10.2 psi or less."
>
> Flame in Freefall challenge statement

At that oxygen level the archive holds **0 solid-fuel runs**, 24 droplet runs and 246 gas-flame runs.



02

Rank: which investigations matter for your habitat

RANK	INVESTIGATION	WHAT BURNED	RUNS	OXYGEN %	NEAR THIS CABIN	SCORE	COMPARE
01	**SAFFIRE-II** PREVENTION Nine smaller samples (silicone, cotton-fiberglass, PMMA, Nomex) burned on Cygnus, each beside its normal-gravity result. Silicone spread little or no flame in weightlessness.	Solid materials	15	21.5 – 22.1	Oxygen in the table runs from 21.5 to 22.1 %, away from this cabin	82	☐
02	**SAFFIRE-I** PREVENTION The first large spacecraft fire experiment: a fabric sheet of 94 by 40.6 cm burned inside an uncrewed Cygnus cargo ship after it left the station, measuring flame growth, oxygen and carbon dioxide.	Solid materials	4	21.5	Oxygen in the table is 21.5 %, away from this cabin	73	☐
03	**FLEX** SUPPRESSION Methanol and heptane droplets burned in many atmospheres to find when a flame cannot survive, and how well carbon dioxide and helium put it out. 274 runs with recorded outcomes, from 12 to 34 % oxygen.	Liquid droplets	274	12 – 34	24 runs at this oxygen level, at another pressure	72	☐
04	**BASS-II** PREVENTION Real spacecraft-type materials (PMMA films, rods and slabs, cotton-fiberglass fabric) burned in a slow airflow at 13.6 to 22.2 % oxygen. It tests whether NASA's normal-gravity screening test is the worst case. Results are in video, still photographs and papers, not in the table.	Solid materials	129	13.6 – 22.2	Oxygen in the table runs from 13.6 to 22.2 %, away from this cabin	67	☐
05	**BASS** PREVENTION SUPPRESSION The first BASS series: cotton-fiberglass fabric, PMMA, Nomex and Ultem burned in opposed and concurrent airflow, plus nitrogen as an extinguishing agent. The table lists only the sample material.	Solid materials	122	–	No oxygen level in the run table	59	☐
06	**ACME BRE** PREVENTION A porous gas burner that imitates burning solids, run in NASA's three candidate cabin atmospheres. Flames in elevated oxygen kept burning after the airflow stopped, while flames at 21 % went out. NASA calls the results preliminary.	Gas burner emulating solids	18	21 – 40	13 test days at this oxygen level, pressure not listed	59	☐

- Pick a **station, spacecraft, Moon base or rover, or Mars base**. All 24 investigations are ranked for it.
- The score has four visible parts: named for fire safety by NASA, fuel close to cabin materials, tested near that cabin's air, data ready to use.
- You can change the weights. Every rule is printed.
- `PREVENTION` `DETECTION` `SUPPRESSION` follow the challenge statement word for word.

03

Summarize and compare: what each one burned, varied and measured

SIDE BY SIDE The same in all of them: how the challenge statement treats it. Rows that match are marked.

Clear

	BASS-II	SAME	FLEX
ROLE IN FIRE SAFETY	Prevention	Detection	Suppression
WHAT BURNED	Solid materials	Smoke from overheated materials	Liquid droplets
FUELS IN THE TABLE	2 cm SIBAL; 1/4" clear rod; 1/4" black rod; 1 cm SIBAL; and 45 more	Teflon; Kapton: same conditions as 1-4; Silicone Rubber: same conditions as 1-4; Lampwick: same conditions as 1-4; and 1 more	Methanol; Heptane
SAMPLE OR BURNER SIZE	–	–	Droplet initial diameter; mm: 1.1 to 4.67 (163 values)
HARDWARE	Microgravity Science Glovebox, SPICE flow duct	Smoke and aerosol instruments	Droplet apparatus (MDCA) in the Combustion Integrated Rack
PLATFORM AND YEARS	Space station, 2010 to 2022	Space station, 2001 to 2008	Space station, 2007 to 2016
OXYGEN, %	13.6 – 22.2	–	12 – 34
PRESSURE, ATM	–	–	0.69 – 3.05
WHAT CHANGED BETWEEN RUNS	Flow restrictor; Fan display; Air display; Radiometer gain (1000 250 50 10); Calibrated initial O2 % by vol; Initial CO2 % by vol; Initial CO (ppm)	Air Flow (cm/s); Aging Duration (minutes)	Ambient pressure; mmHg; O2 in the atmosphere; mole fraction; N2 in the atmosphere; mole fraction; CO2 in the atmosphere; mole fraction; He in the atmosphere; mole fraction

- Each investigation opens to its objective, fuels with sample or burner size, **what changed between runs, what stayed the same** and what was recorded.
- These lists are computed from NASA's run tables, not typed in.
- Tick two or three to see them **side by side**. Matching rows are marked.
- Six hardware families show which experiments can be compared directly.

04

Interpret: grade the evidence for any cabin and list the tests still needed

3 of 6 questions have no flight evidence for this cabin

34 % oxygen, 8.2 psia, lunar gravity. The fire response starts with a step that is unproven in this air: stopping ventilation.

Will the cabin materials burn in this air? D NO FLIGHT EVIDENCE

No selected material was flight-tested at 34 % oxygen. The archive stops at 22.2 %. NASA's guide adds that the ground approval test is not conservative: materials can burn at lower oxygen levels in microgravity than on Earth.

PMMA (acrylic)	D	Tested from 13.6 to 22.2 % oxygen: 11.8 points short of this cabin.
Cotton-fiberglass fabric (SIBAL)	D	Tested from 16.4 to 22.1 % oxygen: 11.9 points short of this cabin.
Nomex	D	Tested from 21 to 22.2 % oxygen: 11.8 points short of this cabin.

Based on F-solid-gap · F-one-g

Does stopping ventilation put the fire out? C INDIRECT EVIDENCE

The evidence points to no. No solid material was tested at this oxygen level, and in the one experiment at elevated oxygen (ACME BRE, a gas burner standing in for solids) flames kept burning with no airflow. Your ventilation speed, 10 cm/s, is in the range NASA's guide calls most flammable (about 10 cm/s).

Based on F-vent · F-vent-speed

Will carbon dioxide put the fire out? D NO FLIGHT EVIDENCE

No flight run with carbon dioxide above 25 % oxygen. Quantities for this cabin cannot be scaled from station values.

Based on F-suppress · F-local-jet

- **Mission check:** set oxygen, pressure, gravity, ventilation, materials and extinguishing agent. Six fire-safety questions are graded A to D by written rules.
- Moon base: **3 of 6 questions have no flight evidence.** Space station: evidence for all six.
- **15 ranked findings** for prevention, detection and suppression, each with evidence, limits, an action and NASA sources.
- Example: in elevated oxygen, flames kept burning after the airflow stopped, the first step of today's fire response.

05

Data for models: every run as CSV, a predictor that knows its limits, results read from NASA video



82 %
right on test days the model never saw

69 %
always guessing the common outcome

- All **986 runs download as CSV**, or only those near your cabin.
- Predictor trained on 261 FLEX droplet runs. It flags extrapolation.
- Flame spread measured from raw BASS-II video: **5 of 5** independent checks pass.

06

Built to be trusted, free to run, open about SI

TRUST

- One script recomputes every number from NASA's raw tables.
- Every quote is matched word for word to its source, or the build stops.
- 44 automated tests recount key facts with independent code.

USE OF SI

- SI (Claude) helped write code and draft summaries, findings and answers.
- Code checks all 81 claims in the answers against NASA records.
- No model runs when you use it: no API, no key, no cost.

NEXT

- Bring the Saffire-IV, -V and -VI and FLEX-2 run data into PSI.
- Read all 244 BASS-II videos into a public results table.
- On Earth the same science already made firefighter gear and smoke detectors better.

NASA data: Physical Sciences Informatics combustion investigations, BASS-II raw video and report, Researcher's Guide to ISS Combustion Science. Research prototype, not a design basis. Not an official NASA product. Slides laid out with SI assistance; pictures are screenshots of the live site.